NEMATODE_OFFICIAL · LABELBOX EXPORT

From LabelBox export to a YOLO demo dataset

1,000 crops · 512×512 px · two classes: nematode-head and cyst

November 2026

The export: 4K microscope frames with box labels

2,772
images, almost all 3840×2160

6,248
bounding boxes, nothing else

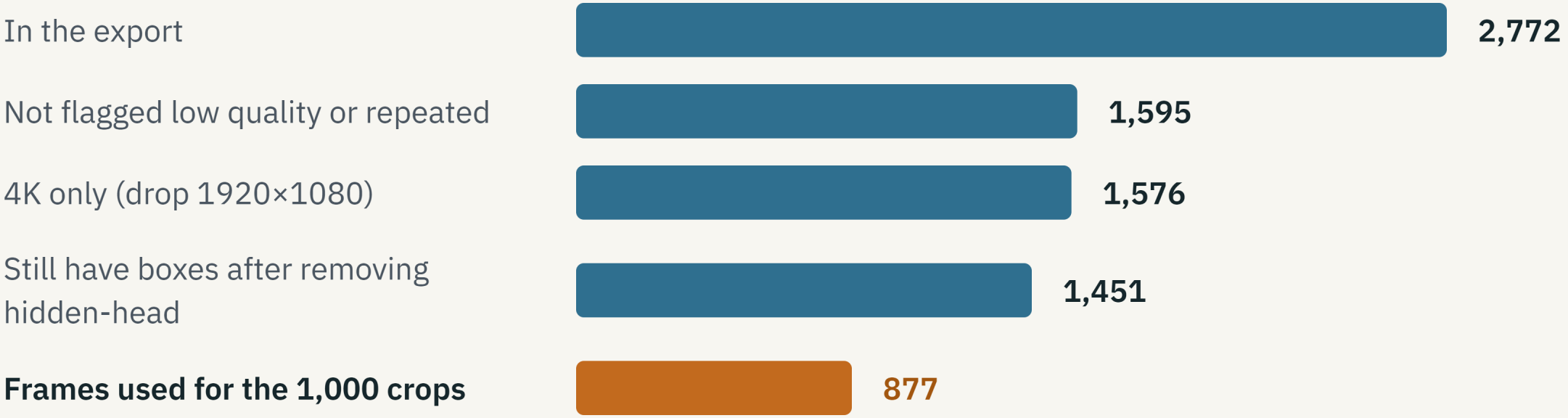
15
original classes

~370
source videos, Jan–Mar 2025

Each box marks **the head of a nematode** or **a whole cyst**.

One image-level question, “how is the image?”, flags frames as Low quality, Repeated, or both.

Filtering: 2,772 images down to 877 source frames



Fifteen original classes become two

0 · nematode-head

Root-knot J2 · SCN J2 · SCN male · lesion · Others · others-lowconfidence · Other-non-plant-parasitic · Unknown

All confidence levels merged

1 · cyst

soybean-cyst-nematode-Cyst, low and high confidence

Rare on purpose: part of the challenge

Removed · hidden-head

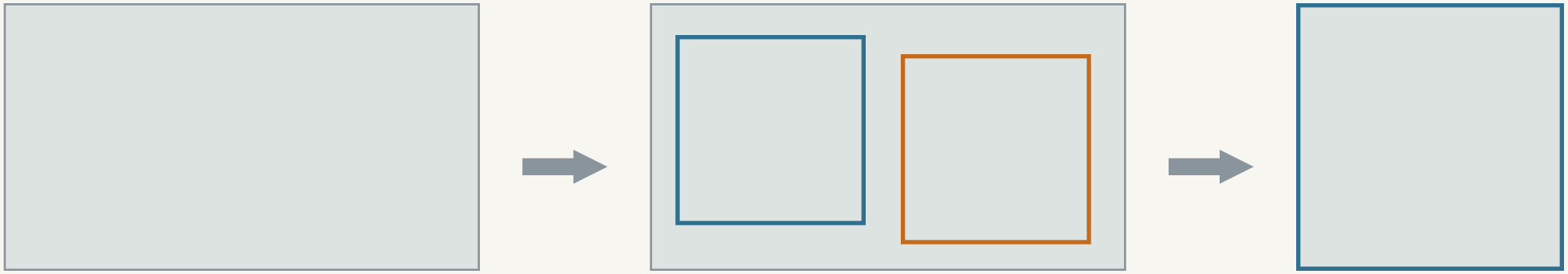
The head is not visible, so the box has no clear target

Shrinking by one third keeps heads visible and cysts whole

Shrink	Frame	Median head	Small heads (5th pct)	Large cysts (95th pct)
1/6	640×360	~15 px	~7 px, too small	~230 px
1/2	1920×1080	~47 px	~20 px	~690 px, won’t fit
1/3 (chosen)	1280×720	~31 px	~13 px	~460 px, fits

Sizes are the long side of each box. A 512×512 crop holds nearly every cyst at 1/3, and heads stay well above YOLO’s small-object limit.

Shrink, then cut 512×512 crops



Original 3840×2160 | Shrunk ÷3 to 1280×720 | 512×512 crop

Every box inside is labeled

Edge-cut boxes are clipped; kept if ≥45% is inside

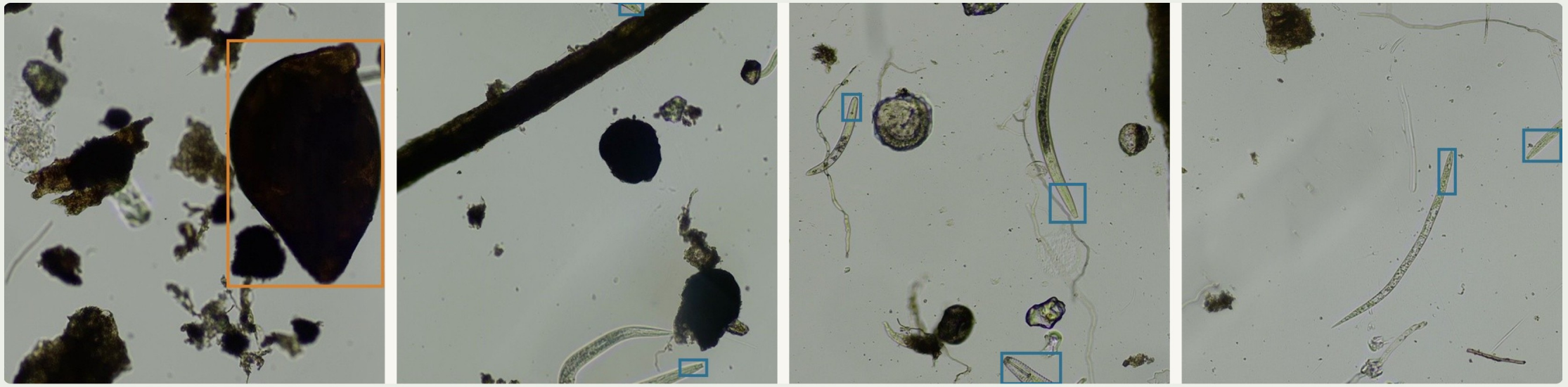
Random offset, never centered

Objects land anywhere in the crop

No stretching

16:9 kept; crops are plain cut-outs

What the crops look like



Orange: cyst, median ~198×219 px **Blue: nematode-head**, median ~26×25 px

1,000 crops, weighted toward rare types

87
cyst crops: every cyst

813
head crops, quotas by √frequency

100
empty crops: no box of any class

Head type behind the crop	Heads available	Crops chosen
Other-non-plant-parasitic	1,618	288
Others	722	192
Root-knot J2	391	140
SCN J2	281	90
Lesion	134	80
Unknown	20	12
SCN male	12	11

60/20/20 split by sample, every type in every split

Crops by type	Train	Val	Test
Cyst	52	17	18
Other-non-plant-parasitic	172	56	60
Others	115	39	38
Root-knot J2	81	30	29
SCN J2	55	18	17
Lesion	47	17	16
Unknown	7	3	2
SCN male	5	4	2
Empty	60	20	20
All crops (sample groups)	594 (185)	204 (70)	202 (65)

Known issues to keep in mind

Worm bodies in empty crops

About 25 of 100 show a body or tail with no head. Good hard negatives, but worth a manual look.

Unlabeled worms

Labels mark heads only; some unboxed worms may show a head nobody labeled.

Two suspicious cyst boxes

About 21 px, head-sized: likely heads mislabeled as cyst.

Expiring image links

LabelBox URLs expire around Nov 14, 2026; rebuilding later needs a new export.

Using and rebuilding the dataset

Train at 512

```
yolo detect train data=nematode_yolo/data.yaml
imgsz=512
```

Detect on full frames

Shrink a 4K frame to 1280×720 and run at imgsz=1280: heads look the same size as in training.

Rebuild from Data-Nema/ (seed 42)

1. **src/select_crops.py**: filter, plan crops, split
2. **src/download_frames.py**: fetch only the frames needed
3. **src/build_dataset.py**: resize, crop, write labels, crops.csv and boxes.csv